Logbook for week beginning 12/10/2026

12/10

Scheduled meeting with Dr. Seeburger to discuss the project for Wednesday 14th.

Was provided with a set of materials to introduce myself into the project:

- El-Badry & Rix, 2022 – arXiv:2206.07723v2
- El-Badry et al., 2024 – arXiv:2405.00089v
- Simon et al., 2026 – DOI: 10.3847/1538-4357/ae7e8a
- Simon et al., 1994 – Bibcode: `1994A&A...281..286S`

I identified that the first question needing to be answered for the project is whether to use UVES, FEROS or a combination of the two, based on the project's description. To do this, I wrote a python script, which used ADQL to query the Gaia DR3 database (El-Badry & Rix, 2022). This returned 23 targets matching the parameters of: that the lower limit on the companion mass exceeds the upper limit on the primary mass, this second star's lower limit mass to be larger than 1.5 but less than 3 solar masses, are tagged as SB1 (only one star's lines are detected) and has a significance over 20 (El-Badry & Rix, 2022). The results of this were then saved to a CSV containing Gaia ID, CMD inferred mass of the primary star, RA and Dec, etc. Using the RA and Dec and a cone radius of 0.1 degree, the UVES and FEROS databases were queried for the data of each target. From this UVES found 6 sources with 49 spectra, and FEROS with 18 sources and 133 spectra. The spectra were then plotted for diagnostic testing, using a simple mean of the flux at every wavelength.

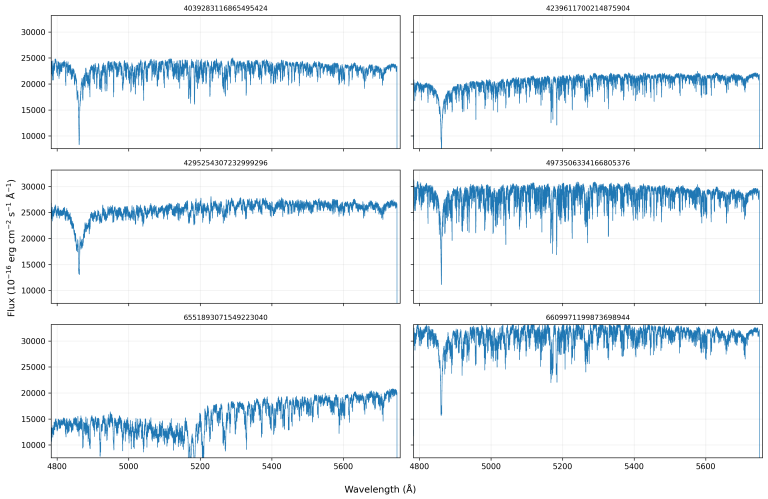


Figure 1: Averaged spectra of all targets identified in UVES database. Plotted until 576 nm where the spectral gap caused by the physical space between the two CCD chips that make up the instrument's red arm detector mosaic occurs (Bagnulo et al., 2003).

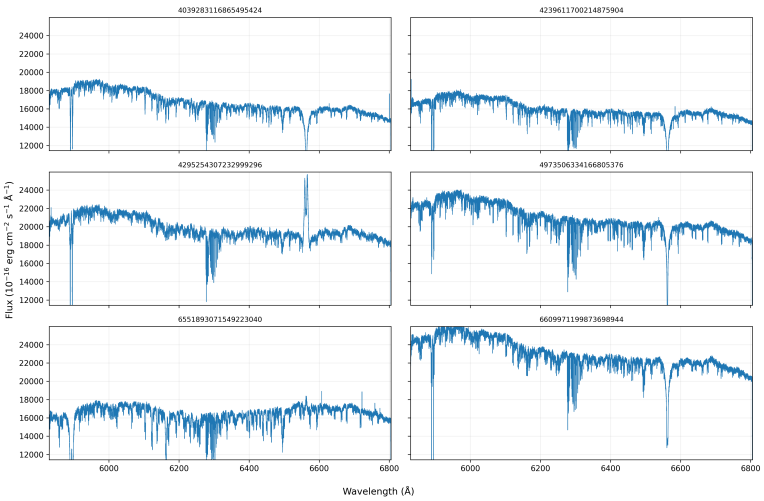


Figure 2: Averaged spectra of all targets identified in UVES database. Plotted from 576 nm.

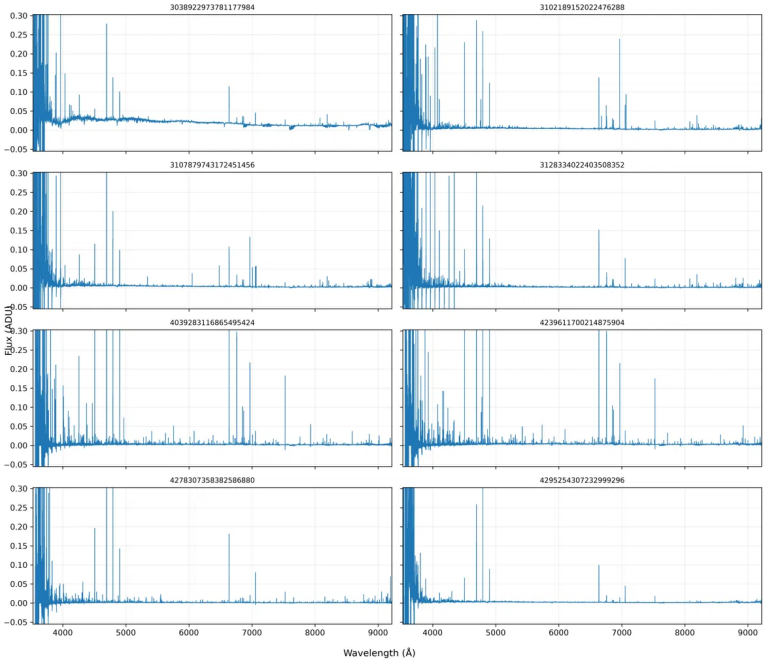


Figure 3: Averaged spectra of eight targets identified in FEROS database. Plotted across the complete band of wavelengths for all these targets. Shows strong noise lines below 380 nm from SNR dropping rapidly (FEROS-II User Manual).

It is worth noting that the data from UVES in the fits files are already flux calibrated, but the FEROS data is not.

I then created this table to better understand the benefits and costs of both instruments.

	FEROS	UVES	Advantage
Data	18 Sources, 133 Spectra	6 Sources, 49 Spectra	FEROS
Resolution	48,000	66,320	UVES
Coverage	350–920 nm	473–684 nm	FEROS
SNR (Median)	20	67	UVES
Artifacts/Noise	Many	Some	UVES
Flux calibration	No	Yes	UVES
Targeting	Issues	No issues	UVES
Consistency	Varied	Uniform	UVES
Time span	60–165 d to 697–900 d	60–160 d	FEROS

Notes:

- Coverage of UVES contains spectral gap discussed in Figure 1 caption.
- FEROS spectra below 380 nm heavily contaminated and would need to be cleaned first. Many artifact lines also present.
- Targets in FEROS database contain observations over a decade with many different programmes. All UVES are from Dr. Rhys Seeburger's programme(s).
- Target 5610703902634458112 may be a different target. Further research needed if intending to use FEROS.
- Significance defined as RV semi-amplitude divided by its uncertainty (El-Badry & Rix, 2022).

It is therefore that I have concluded the most optimal route is the use the UVES data as the primary source and to use FEROS for any targets not in the UVES database or as additional data if time allows.

Programs created: `ADQL_test.py` (pyvo, csv, pathlib, matplotlib, requests, astropy, os, numpy, astroquery).

Files generated: `Gaia_Data_DR3.csv`, `FEROS_data.csv`, `UVES_data.csv`, combined grid spectra, combined target spectra, individual fits spectra.

13/10

Finished reading El-Badry & Rix, 2022.
Notes on paper:

I found the methodology of the study useful, particularly for ADQL (2. Sample Selection) which I have not used before and the definition and use of "significance" in the Gaia catalogue.

The spectral analysis portion of the paper goes into good detail on the features to look for, such as double-peaked emission in Hα and its potential origin in the accretion disk.

While the main goal of the paper was to study inferred binary systems with $M_{\rm companion} \geq 3\,M_\odot$ compared to this projects $1.5\,M_\odot \leq M_{\rm companion} \leq 3\,M_\odot$, there is much in the analysis which is relevant. For example, the study found that after modelling the systems as two luminous stars, the actual companions came out as $M_{\rm companion} = 1.9\,M_\odot - 2.4\,M_\odot$. This suggests that some of the 23 targets may have lower true masses than inferred. In the paper this happened because the companion was a luminous star, so the same effect would make those targets imposters rather than neutron star candidates.

The specifics of the results analysis aren't relevant due to the difference in target, however.

To recreate the bottom-right section of Figure 1 from Simon et al., 2026, I produced a Colour-Magnitude Diagram (CMD) using a random sample of 100,000 targets from the Gaia DR3 database. On this, our 18 candidates with spectra were then overlaid on top.

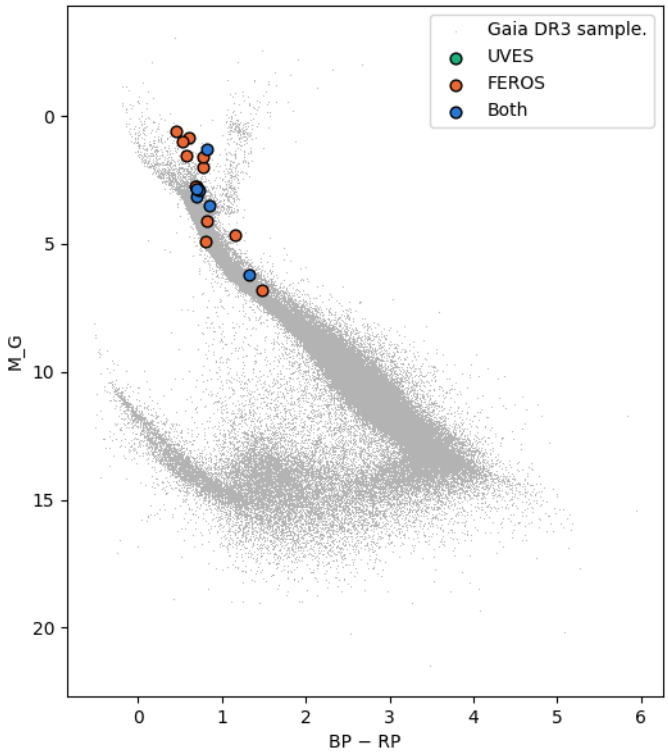


Figure 4: Colour magnitude diagram of 100,000 background targets from the Gaia DR3 database with candidates described above overlaid.

This indicates that the 18 candidates with spectra lie on, or just above, the upper main sequence, with none on the giant branch. Some sit 2–3 magnitudes above the main sequence at their colour, consistent with either evolution into the Hertzsprung gap or interstellar reddening, which has not yet been corrected. Regardless, it supports the assumption that the primary star in each of these candidates is a main sequence or slightly evolved star.

The next step is to calibrate the files. Looking at the headers of the fits files shows that the UVES observations are topocentric and all of the FEROS files are barycentric. Therefore, barycentric corrections need to be performed on the UVES data. To do this, I first checked the header of the fits file for its `ESO QC VRAD BARYCOR` value, this gives the Earth's radial velocity around the solar system's barycentre relative to the target (considering the Earth's rotation too). I could then use $1 + \frac{v_{\text{radial}}}{c}$ to correct the given wavelength into a barycentric one.

After processing all topocentric files into barycentric ones, the spectra were plotted again.

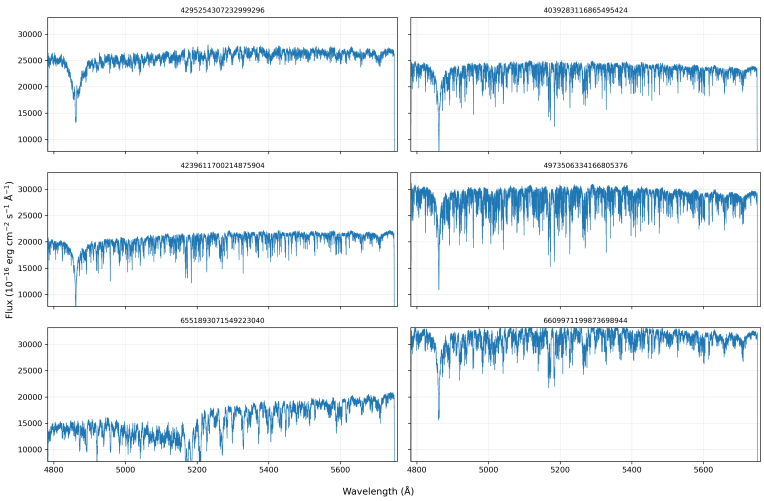


Figure 5: Averaged spectra, with barycentric corrections, of all targets identified in UVES database. Plotted until 576 nm.

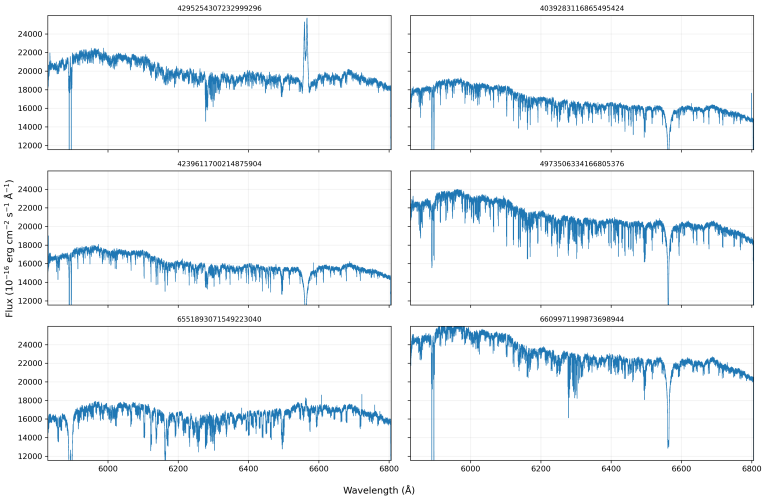


Figure 6: Averaged spectra, with barycentric corrections, of all targets identified in UVES database. Plotted from 576 nm.

Figure 5 and 6 show clear signs of the results of the barycentric corrections. For example, the trough at approximately 627 nm for 4973506334166805376 in the non-barycentric spectra peaks at approximately a flux of 13000. Whereas, in the barycentric plot, the peak of the trough is approximately 17000 and much less dramatic. Now that the barycentric corrections have been made, the next step was to normalise the spectra. My first attempt of this was to implement the standard normalisation equation $\frac{Flux - Flux_{\min}}{Flux_{\max - Flux_{\min}}}$. This, however, did not consider the slope, chip step and peaks due to cosmic rays, all of which bias the results. Therefore, I rewrote the function to normalise the spectra by fitting a 5th order polynomial to estimate the continuum of each chip separately (split at 5760Å, excluding zero-flux pixels), keeping only the pixels above the fit (so the absorption lines don't drag it down), refitting to only those pixels, then dividing the flux by this second fit. This produced normalised spectra for UVES and FEROS.

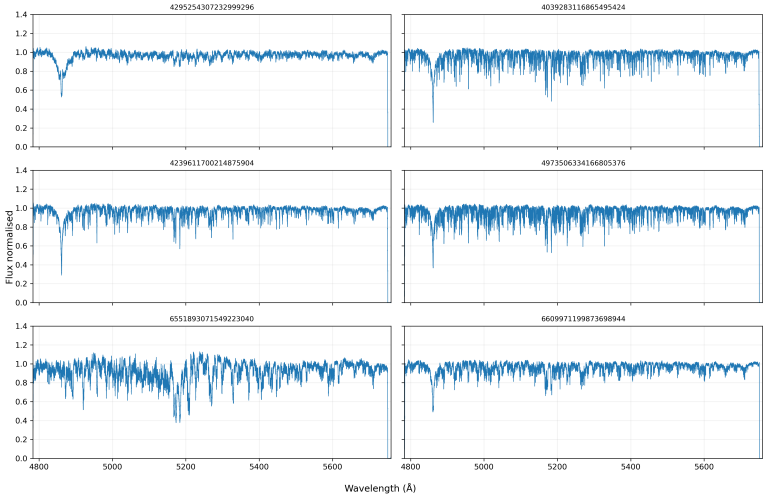


Figure 7: Normalised average spectra, with barycentric corrections, of all targets identified in UVES database. Plotted until 576 nm.

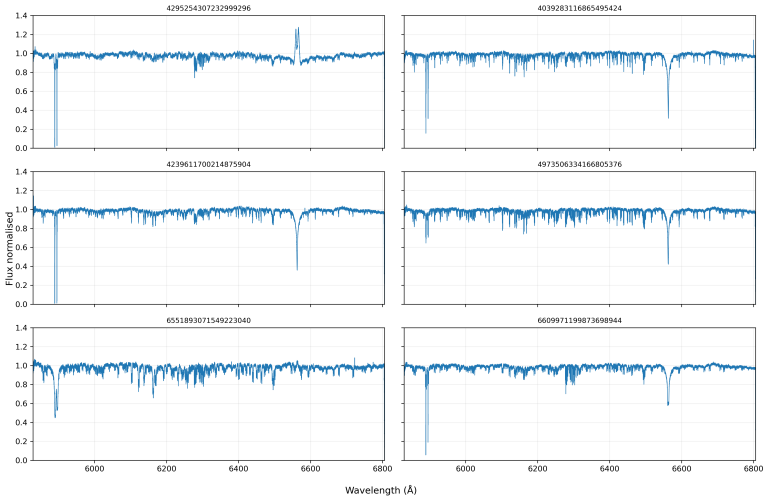


Figure 8: Normalised average spectra, with barycentric corrections, of all targets identified in UVES database. Plotted from 576 nm.

Using these, I could attempt to identify prominent absorption and emission lines visually. Hβ absorption line at approximately 486nm is visible in all spectra except for 6551893071549223040 (though it may be present and blends in with all other absorption lines). The Hα absorption line at approximately 656nm is visible in all spectra except 6551893071549223040 and 4295254307232999296, where it is present as a double-peaked emission line in the latter. This double-peaked emission line is potentially from the accretion disk, caused by the difference in rotational velocities (El-Badry & Rix, 2022). However, further processing is needed before these spectra can be used for proper scientific analysis. Further parameters were added to the ADQL Gaia query: period, eccentricity, semi amplitude, center of mass velocity, RUWE, radial velocity and goodness of fit. From this, I could produce a complete table.

Table 1: Astrometric and spectroscopic parameters of the binary star candidates.

Source ID	RA (deg)	Dec (deg)	m_1 ($M_\odot$)	$m_{1,\mathrm{lower}}$ ($M_\odot$)	$m_{1,\mathrm{upper}}$ ($M_\odot$)	m_2 ($M_\odot$)	$m_{2,\mathrm{lower}}$ ($M_\odot$)	$m_{2,\mathrm{upper}}$ ($M_\odot$)	NSS Type	Significance	Period (days)	Eccentricity	K_1 ($\mathrm{km\,s^{-1}}$)	v_{cm} ($\mathrm{km\,s^{-1}}$)	RUWE	RV ($\mathrm{km\,s^{-1}}$)	GoF
5558841347897409408	104.5912	-44.5106	2.2068	2.0299	2.3947	–	2.5099	–	SB1	49.2289	12.8198	0.0678	84.1299	12.2771	0.9607	-2.0303	-1.9110
6609971199873698944	343.2005	-27.2347	1.2668	1.1886	1.3390	–	1.9387	–	SB1	36.4726	18.3423	0.1939	75.6583	30.6227	0.9025	18.8124	1.9993
4728162202895516288	42.7545	-56.5590	1.5722	1.4862	1.6566	–	1.8112	–	SB1	73.5841	114.7635	0.3804	38.8210	20.7782	1.6525	22.2368	2.9802
4341334461557059328	253.0953	-5.8243	0.7223	0.6688	0.7758	–	1.8107	–	SB1	21.7076	353.7105	0.3796	33.4587	-21.4053	5.5775	-36.6136	2.2993
1702370142434513152	240.8916	72.6590	1.0748	1.0153	1.1330	2.5271	1.5409	3.4700	SB1	32.0692	101.1090	0.1087	22.5159	9.6143	2.5015	10.1500	2.1583
4295254307232999296	293.6932	7.8418	2.0948	2.0225	2.1518	–	2.6222	–	SB1	29.5598	10.8328	0.0461	93.2564	5.7739	1.0419	26.6144	-0.2001
5610703902634458112	105.0774	-27.1662	2.0456	1.9772	2.1109	–	2.7490	–	SB1	48.1408	1003.2501	0.0211	21.0175	19.6473	2.2270	20.3391	0.1261
3107879743172451456	106.3760	-3.3923	1.8508	1.7883	1.9082	–	2.6413	–	SB1	21.3142	27.3049	0.0268	72.5399	29.6273	0.9810	34.9536	0.0765
4973506334166805376	1.0514	-50.9169	1.3344	1.2553	1.3921	–	1.5461	–	SB1	98.3969	57.5235	0.2820	44.7704	2.8078	1.7893	18.4837	0.5798
209136938490844928	80.9814	45.9052	2.0790	2.0099	2.1481	–	2.5730	–	SB1	30.3899	14.2915	0.0429	83.7893	-42.4742	0.9865	-6.8384	2.0289
3038922973781177984	119.6007	-9.9268	1.3728	1.2984	1.4381	–	1.5419	–	SB1	103.8494	58.8274	0.5258	49.5511	16.9306	2.6325	8.7203	0.8456
5461264982333157888	155.9023	-30.5772	0.9107	0.8611	0.9657	1.6475	1.5132	1.7950	SB1	21.6735	762.7745	0.3162	21.4076	15.9051	3.6434	16.5369	2.0763
4039283116865495424	273.1124	-34.6049	1.4224	1.3616	1.4784	–	1.7919	–	SB1	20.9381	79.5488	0.1277	43.3654	-6.4591	0.6583	27.3359	2.0963
4345739070777246080	244.6805	-8.3813	1.9147	1.8456	1.9805	–	2.0040	–	SB1	29.6918	6.1961	0.0959	97.4694	107.5146	1.1989	67.5146	1.1769
2220072241734878720	330.7108	67.9182	1.5593	1.5029	1.6157	–	1.6387	–	SB1	21.3815	331.1910	0.1791	24.8503	-3.9673	0.9615	0.7145	2.0284
4278307358382586880	275.6243	4.0058	0.9773	0.9194	1.0366	–	1.6241	–	SB1	29.4791	11.9627	0.1741	84.2552	-7.9755	1.1677	30.1902	-0.8035
1842500631652332544	312.2268	24.0314	1.8331	1.7792	1.8922	–	2.2961	–	SB1	22.0077	4.5800	0.1054	120.1778	-4.5685	1.0549	8.3671	0.1494
4239611700214875904	295.6654	0.3377	1.3327	1.2669	1.3841	–	2.3952	–	SB1	23.5991	56.1241	0.4041	62.9487	-9.8261	1.1089	-1.4796	2.4260
3102189152022476288	102.1684	-4.8640	1.0083	0.9504	1.0675	–	1.6273	–	SB1	53.5900	6.0855	0.1528	103.4855	-7.1706	0.9564	-35.6281	2.1905
3128334022403508352	105.3557	3.8860	1.4431	1.3852	1.4978	–	1.5015	–	SB1	20.9559	7.1430	0.4599	94.6182	59.6592	0.9993	33.35098	2.1101
5780828175702576384	245.6773	-76.3603	2.0775	2.0097	2.1386	–	2.3187	–	SB1	80.7530	10.7496	0.0488	84.8460	-7.3345	0.9488	-18.0976	0.0487
6475655404885617920	308.6084	-50.6256	0.9839	0.9289	1.0356	–	1.5125	–	SB1	20.4828	46.0947	0.1013	51.4143	29.7381	2.9373	41.5155	1.3857
6551893071549223040	349.4574	-36.4026	0.8050	0.7584	0.8566	–	1.6830	–	SB1	36.3350	79.3952	0.0925	46.9704	68.0310	1.0350	63.9081	2.3394

RUWE is defined as the reduced unit weight error. Goodness of fit is suggested to be limited at no more than 4 (Simon et al., 2026).

I then implemented the python function `find_peaks` from the package `scipy.signal` to identify each absorption peak in each spectrum, where a peak is defined as any point at least 10% below the surrounding flux. These peak wavelengths were then saved to a CSV.

Programs created: `normalised_spectra` function added to `ADQL_test.py`. Files generated: Updated `Gaia_Data_DR3.csv`, `UVES_peaks.csv`, normalised spectra grids for both UVES and FEROS.

FEROS normalised spectra not discussed as not intended for use and requires much cleaning before any real analysis can be performed.

14/10

Finished reading Simon et al., 2026. The paper is a study of candidate compact binaries across MIKE, APF, IMACS and FEROS observations. The paper notes that two confirmed black holes and a number of neutron stars have already been found in the Gaia DR3 catalogue, which gives credibility to the project. It confirmed the method for radial velocity measurements. It also highlighted the possibility of contamination from other sources and that each target must be checked in case it is a known source, such as a Be star, which could cause contamination. The paper also introduces a number of other variables to query the Gaia database for: significance, period, eccentricity, velocity semi-amplitude, center-of-mass velocity, reduced unit weight error, radial velocity and goodness-of-fit. It uses a colour-magnitude diagram to demonstrate the evolutionary stage of the primary star, this was de-reddened using a 3D dust map from Wang et al., 2025. It later uses Bayestar extinctions in its isochrone fits for the stellar masses. For its MIKE observations, the paper uses a method to obtain the radial velocity by applying a χ^2 fit order by order, using the radial velocity standard star HD 126053 as a template, and excluded the orders with strong telluric absorption and those covering the Na I D lines. The paper also suggested that spectra for hotter stars may be inaccurate for radial velocity measurements, though suggests a method to do this if needed. It uses the method detailed in El-Badry et al., 2024c for FEROS radial velocity measurements. To obtain spectral-based inferences [Fe/H] and micro-turbulent velocities, they used equivalent widths of Fe I and Fe II lines, measured using Gaussian profiles with the `Spectroscopy Made Harder` program. They also note that the RUWE value is elevated for stars that exhibit astrometric orbital motion and orbits with smaller eccentricities and inclinations. To obtain the final orbital parameters, they used a program called `TheJoker` and `emcee`. The paper notes that for some cases, radial velocity measurements were only able to be made using the Hβ absorption line and that faint sources and ones with broad lines can lead to erroneous classification. Finally, the paper finds that of their 23 objects, 18 were binary systems, though only Gaia BH1, BH2 and NS1 contain neutron stars or black holes. Four were very hot rapidly rotating stars, and the last star, the faintest, showed no RV variation. Aside from BH1, BH2 and NS1, the Gaia orbits for all companions above $2M_\odot$ are incorrect. The paper suggests the following quality cuts: significance > 10, goodness-of-fit < 4, which all our targets pass.

I had a very successful meeting with Dr. Seeburger where we discussed the specifics of the project in detail. What was particularly useful was the discussion of the cross-correlation methodology. I was informed that the method to do this was to take synthetic template spectra and normalise them, then to interpolate by creating even spacing in the bins of log λ plots. Then use this to degrade the resolution. This can then be used to interpolate our plots. To do this, the synthetic spectra would need to be the same resolution of our spectra, this being 66,320 for the UVES observations. Dr. Seeburger was also very helpful with answering my questions, such as that the data to be used is the six candidates with spectra in both UVES and FEROS and that these datasets are his and haven't been properly analysed before, what python packages to use (such as TheJoker and his own spectral disentanglement script). Dr. Seeburger also helped with the initial analysis

of some targets, explaining the possible causes behind the double peaked Hα emission and the Na D lines. Dr. Seeburger also helped me understand the scope of the project, its overall aims and the amount of latitude I have personally with the direction of the project. Overall, the meeting was incredibly informative in helping me understand the theory behind the project and the practical next steps.

Agreed plan until next meeting: Dr. Seeburger advised an initial exploratory phase, producing and inspecting spectra of the targets to become familiar with the data.

After the meeting, I decided a few things. First, UVES data would be used first as Dr. Seeburger was still in the process of processing the FEROS data.

Secondly, I decided to change the method of how the barycentric corrections and normalisation worked. For barycentric corrections, this was done to increase the accuracy of the radial velocity calculation, rather than relying on the automatic UVES header value. Fortunately, Astropy has a function which is able to perform barycentric corrections using the RA and DEC of the target, the time of the observation and the VLT's location (which is where UVES is). For the new normalisation method, Astropy also contains a modelling, fitting and sigma clip function. These were used to update the normalisation function. To do this, I used a least-squares fitter, a sigma clip that removes all points that are more than 1.5σ below or 3σ above. This was fitted on a 10th degree Chebyshev polynomial and returns the fitted model. This reproduced the two normalised plots. The new barycentering method has a mean velocity difference from the fits header values of 0.104 $\mathrm{km\,s^{-1}}$ with the new method generally finding larger values.

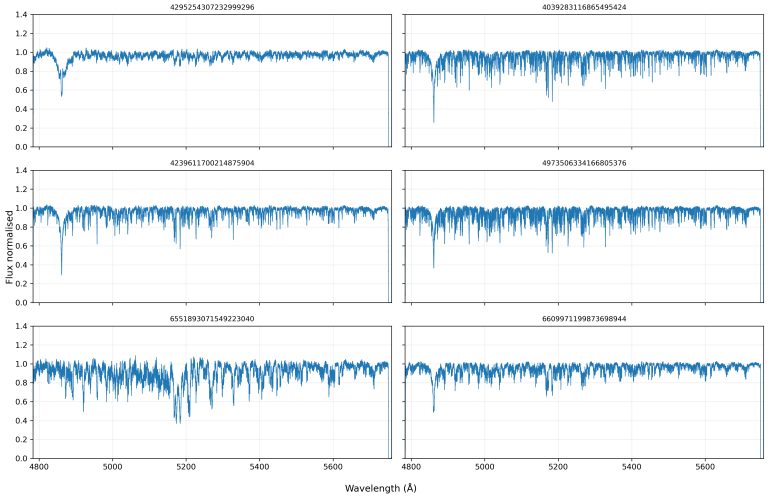


Figure 9: Normalised spectra overlaying all observations, with barycentric corrections, using the new methods for all targets identified in UVES database. Plotted until 576 nm.

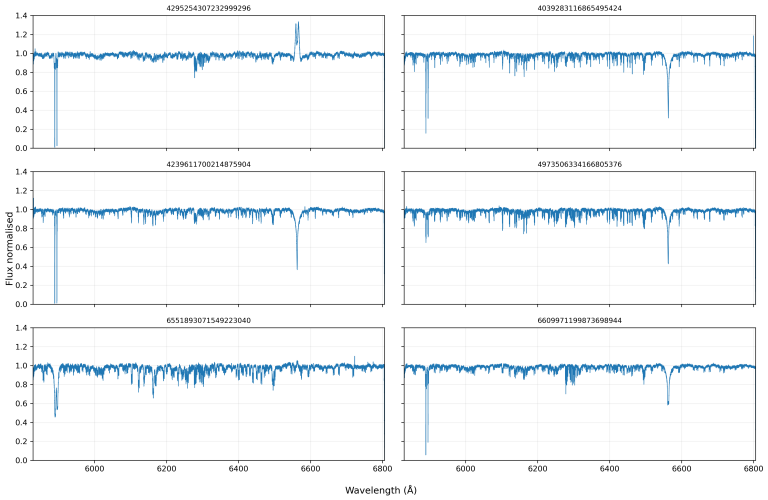


Figure 10: Normalised spectra overlaying all observations, with barycentric corrections, using the new methods for all targets identified in UVES database. Plotted after 576 nm.

As discussed with Dr. Seeburger during the meeting, the Hα emission in 4295254307232999296 and Na D lines across

all spectra are of particular interest.

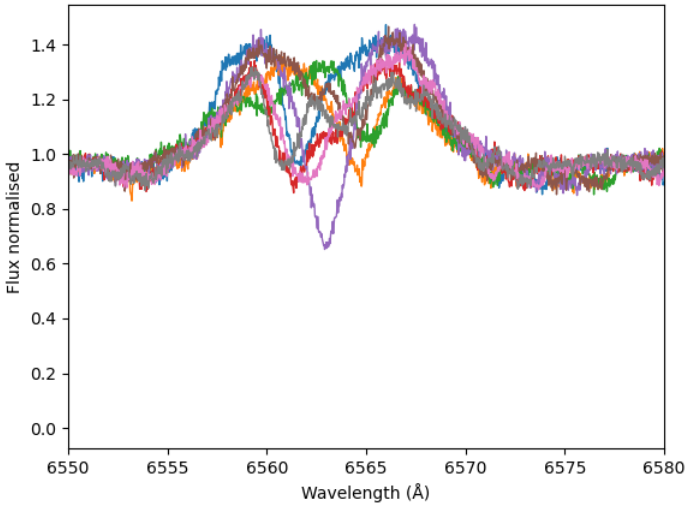


Figure 11: Normalised spectra with all observations overlaid, with barycentric corrections of 4295254307232999296 between 655nm and 658nm. Showing Hα line.

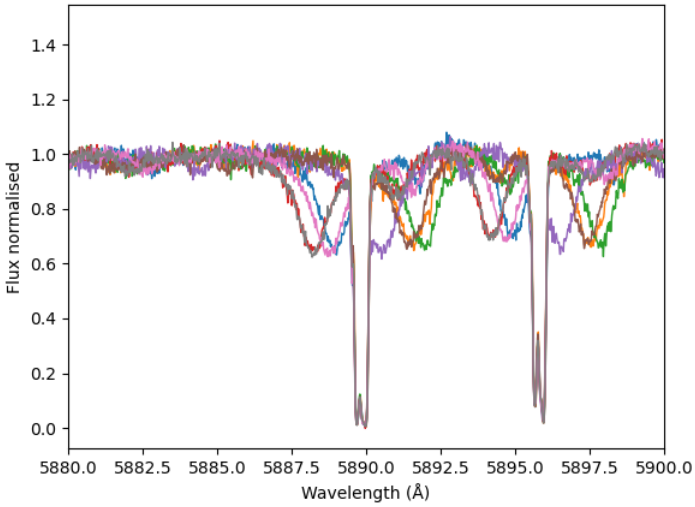


Figure 12: Normalised spectra with all observations overlaid, with barycentric corrections of 4295254307232999296 between 588nm and 590nm. Showing Na D lines.

Figure 11 shows the Hα lines. As expected, the emission lines are smeared across the waveband. In absorption lines, the peaks move as the smaller mass star orbits around its companion, this causes a consistent shifting in the wavelength of the peak. For interstellar lines, these peaks are fixed at a wavelength as the medium causing the absorption is relatively stationary on our time scales. Therefore, one potential cause of these emission lines is their origin in the accretion disk (El-Badry & Rix, 2022).

Figure 12 shows the absorption likely due to the Milky Way (Poznanski et al., 2012). The consistency of these lines across all observations of the target tells us that they are not a property of the target, but come from the general interstellar medium of the Milky Way, which lies along the line of sight to every target rather than being a single cloud. However, inconsistencies in peak depth across different targets indicates the need for further investigation.

Code: Updated Barycentric corrections and normalisation code. Created `inspector` function to observe specific wavebands of specific spectra.

Files created: Updated UVES spectra for all target and grids using new barycentric corrections and normalisation. Figures of Hα and Na D lines in 4295254307232999296.

15/10

Now that I have added reasonable barycentric corrections and normalisation functions, I can begin to study these spectra for specific absorption lines. For this, I have used the Fraunhofer lines (Wikipedia, as the website that hosts the paper isn't letting me access it even with my university login) as the comparison absorption lines. Using 4239611700214875904 as the example, I produced the following figures.

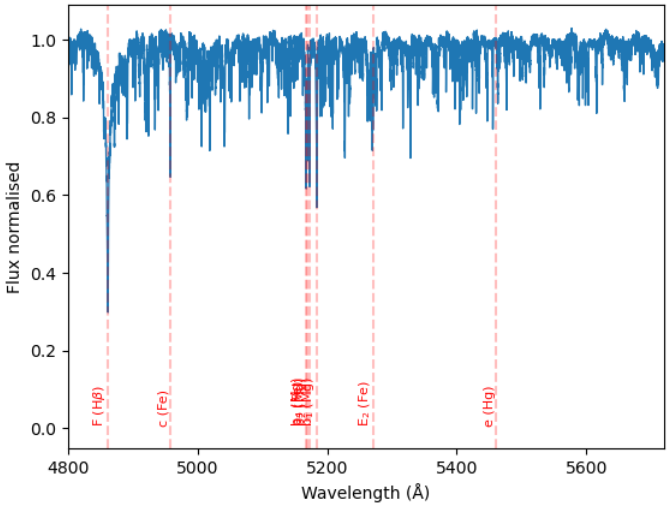


Figure 13: Normalised average spectra, with barycentric corrections of 4239611700214875904, plotted between 480nm and 572nm with spectral absorption lines marked.

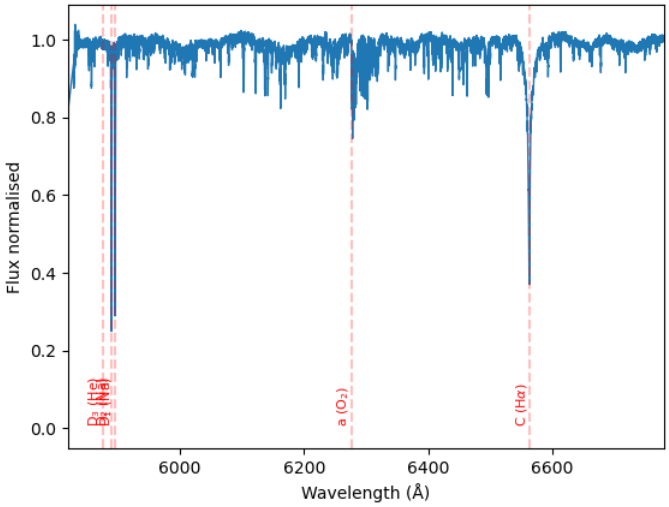


Figure 14: Normalised average spectra, with barycentric corrections of 4239611700214875904, plotted between 582nm and 678nm with spectral absorption lines marked.

Figure 13 shows absorption lines in nearly all the main Fraunhofer lines, except the Hg line. This is expected, as the Hg line is a laboratory emission line from mercury lamps, used as a reference wavelength (Sansonetti et al., 2010), rather than a feature of stellar spectra. Therefore, I discarded this line.

Figure 14 shows absorption lines in nearly all the Fraunhofer lines too. This time the D_3 Helium line is not present. This suggests that the primary star of 4239611700214875904 may be cooler than a B-class star (Gridihar, 2010). This figure also introduces a new interesting property of the spectra, the telluric O_2 line. This originates from absorption due to molecular oxygen in Earth's atmosphere (Wang et al., 2022).

I then produced the remaining Figure 1 plots from Simon et al., 2026, with our candidates overlaid on the Gaia binary population, with a total of 100,000 background stars.

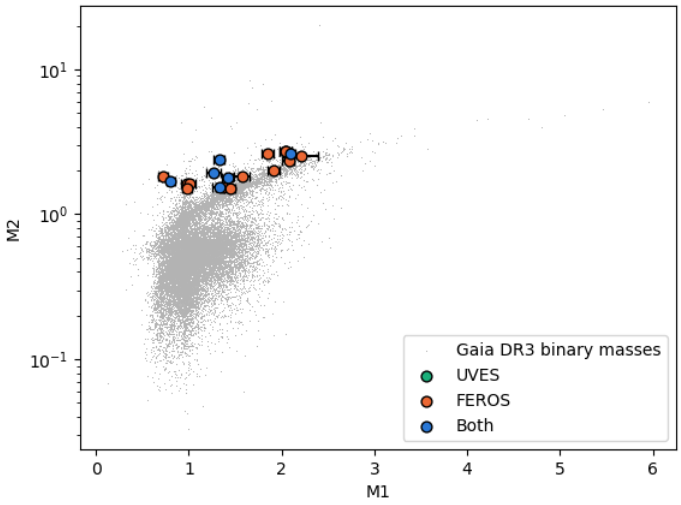


Figure 15: Secondary-Primary mass relation where M2 is the lower bound mass estimate of the secondary star. Shows that our candidates have a higher ratio between secondary and primary masses than the other binaries the catalogue. Infers that the secondary mass is larger than that of a standard main sequence binary, consistent with a compact object.

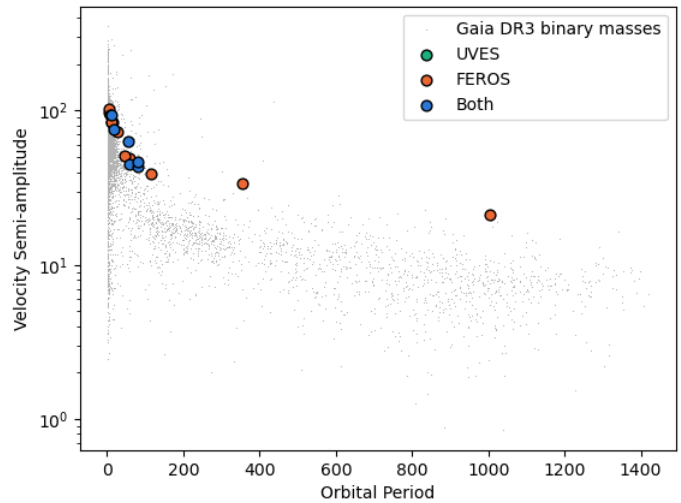


Figure 16: Velocity semi-amplitude-Orbital period relation. Shows that our candidates have a higher velocity semi-amplitude at the same orbital period than that of the binary catalogue average. This infers that the primary star is experiencing a stronger gravitational force than expected with a main sequence star.

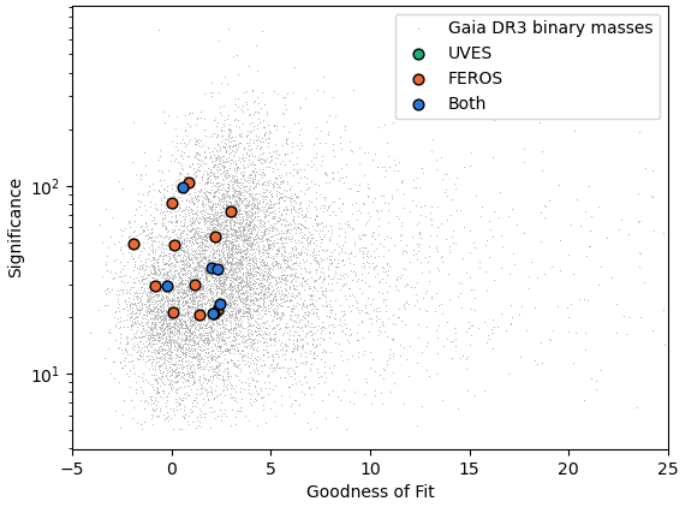


Figure 17: Significance-Goodness-of-fit relation. Shows that our candidates fit consistently within the general binary population. This figure was originally incorrect as the Gaia ADQL call did not include `nss_solution_type = 'SB1'`, rerun maxed out at 9296 background stars.

Code: Changed the CMD plotting code to a general `population` one. Added `spectral_lines` to show the absorption lines on the UVES spectra.

Files produced: Spectra with chosen absorption lines mark on them. Mass-Mass relation, K_1-P relation, Significance-GoF relation.

17/10

As I had largely exhausted the visual checks I could perform on each spectrum, I began working toward setting up cross-correlation. I did not want to implement any systems before meeting with Dr. Seeburger again to understand what the full cross-correlation would involve. So, I decided to try and follow a method as from a published paper, David et al., 2014, as a test run.

First, I had to satisfy the six assumptions about the input data. Assumption 1 (observed spectra have been prepared) had already been satisfied, and assumption 2 (random flux errors statistically independent) was taken as held. Assumption 3 stated that the probability density function (PDF) of the random flux errors is Gaussian with known dispersion. To test this, I implemented a new `error_pdf` function. This function took each individual spectrum, looking specifically at relatively flat continuum regions. For this I use 540–545 nm and 605–610 nm for the blue and red regions respectively. A first degree polynomial was then fitted to each region using $F_{\rm fit}(\lambda) = a\lambda + b$, where $F_{\rm fit}$ is the fitted flux, λ is the wavelength, a is the gradient and b is the intercept. The flux residuals were then normalised by their uncertainties, $r_i = \frac{F_i - F_{{\rm fit},i}}{\sigma_i}$. The standard deviation of these residuals is $s_r = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(r_i - \bar{r})^2}$, where $\bar{r} = \frac{1}{N}\sum_{i=1}^{N} r_i$. The difference between adjacent flux pixels was calculated and normalised, $z_i = \frac{F_{i+1} - F_i}{\sqrt{2}\,\sigma_i}$, where the factor $\sqrt{2}$ accounts for the difference of two pixels with similar uncertainties ($\sigma_{i+1} \approx \sigma_i$). The standard deviation of the normalised pixel differences is $s_z = {\rm std}(z_i)$. The distributions of z_i for the blue and red regions are compared with the standard normal probability density function $p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$, and with a Gaussian distribution of mean zero and width equal to the measured s_z, $p_s(x) = \frac{1}{s\sqrt{2\pi}} \exp\left(-\frac{x^2}{2s^2}\right)$ with $s = s_z$. The mean and standard deviation of s across the M spectra of each target are then $\bar{s} = \frac{1}{M}\sum_{j=1}^{M} s_j$ and $S = \sqrt{\frac{1}{M}\sum_{j=1}^{M}(s_j - \bar{s})^2}$, where s_j is s_r or s_z for spectrum j.

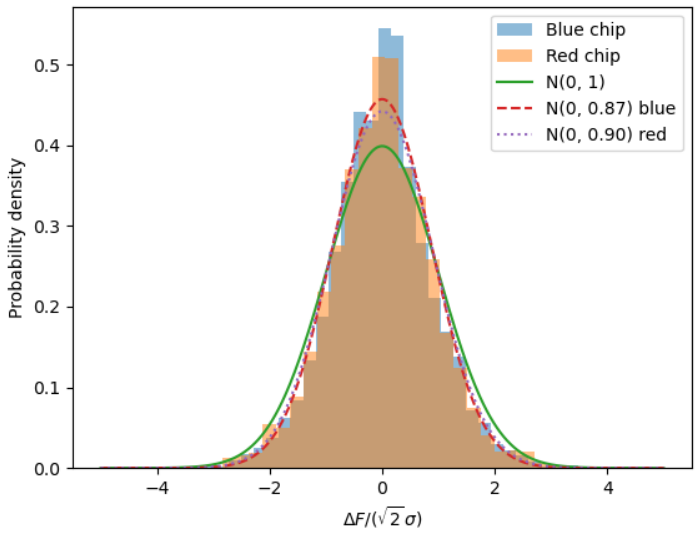


Figure 18: Example PDF plot for observation ADP.2025-05-14T09-09-28.916 of candidate 4039283116865495424.

From this, I produced the following table.

Table 2: Mean and standard deviation across epochs ($\bar{s} \pm S$) of the normalised residual width s_r and the normalised pixel-difference width s_z for each target, in the blue (540–545 nm) and red (605–610 nm) regions.

Gaia ID	s_r (blue)	s_r (red)	s_z (blue)	s_z (red)
4295254307232999296	4.0 ± 0.5	2.1 ± 0.2	0.89 ± 0.02	0.91 ± 0.02
4039283116865495424	7.8 ± 1.2	2.7 ± 0.3	0.96 ± 0.03	0.91 ± 0.04
4239611700214875904	5.9 ± 1.3	2.0 ± 0.2	0.98 ± 0.03	0.92 ± 0.03
4973506334166805376	11.8 ± 2.3	3.8 ± 0.6	1.35 ± 0.15	1.02 ± 0.05
6551893071549223040	8.9 ± 2.3	3.4 ± 0.6	0.94 ± 0.04	0.94 ± 0.04
6609971199873698944	6.1 ± 1.4	2.3 ± 0.4	0.92 ± 0.03	0.94 ± 0.04

As shown in the table, the residual widths, s_r are well above 1 for every target, and approximately 2-3 times larger in blue than in red. This is likely due to the fact that neither region is free of stellar and interstellar lines. Therefore, s_r is measuring spectral line features rather than noise. I, therefore, switched to a pixel-difference, s_z, approach. This approach gives an approximate general range of ≈ 0.89–0.98 in both regions for five of the six targets, which is generally consistent across each observation. This shows that uncertainties describe the noise well and that the distribution is close to Gaussian and therefore satisfying assumption 3. The one exception to this is 4973506334166805376, which is above the average of the targets. My best guess for this cause is that the absorption lines for this target are sharper than that of the other targets, causing the pixel-difference approach to bias, though I can not say that with much confidence. It is worth noting that inconsistency is conserved when shortening the wavelength range to blue (541–542 nm) and red (607–608 nm) regions.

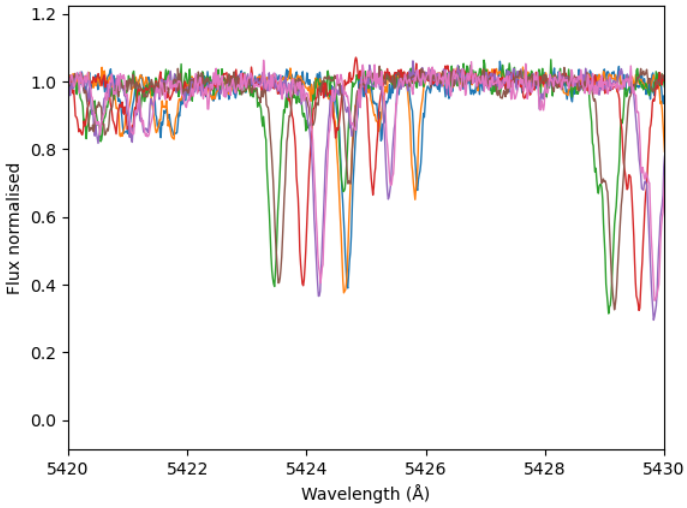


Figure 19: Barycentric and normalised plot of 4973506334166805376 from 542nm to 543nm.

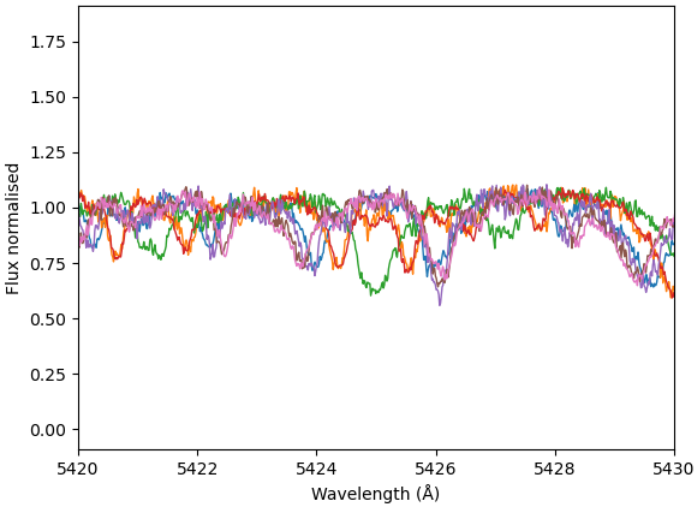


Figure 20: Barycentric and normalised plot of 6551893071549223040 from 542nm to 543nm.

These plots show that 4973506334166805376's absorption lines are deeper and thinner, this would cause the pixel-difference method to bias against it. For now I noted this and concluded that assumption 3 has been satisfied.

I then moved to assumption 4. This states that cosmic rays and CCD blemishes have been flagged and removed. For this, I applied a simple cut for all detections above 2 standard deviations above the median of the continuum. I then ran this through all spectra and removed a total of 145 pixels from all spectra.

I then moved to assumption 5. This states that all spectral non-stellar features should be identified and masked. The initial attempt to do this involved using a known interstellar absorption table from Hobbs et al., 2008. A function was then developed to check all the absorption peaks from this table against the documented peaks from each spectrum. These spectral peaks had been previously flagged using the `find_peaks` function from `scipy` and saved to a CSV (this method flags all dips in the spectra of more than 10% relative to the surrounding flux). If the function found a match between the catalogue and spectral peaks CSV (within 0.05Å, though this was later changed to 0.3Å), it was flagged and saved to a separate CSV. After this, a new `.fits` file was generated with these flagged wavelengths set to a normalised value of 1 (using $\lambda \pm (FWHM/2 + 0.2\text{Å})$, where 0.2Å is the padding) for each existing `_bary.fits` file. However, this method was also incorrectly flagging stellar lines. As a result, this method was switched to one which would look at the absorption peaks from the `find_peaks` method and check them against the other observations for the same candidate. If these peaks varied in wavelength across observations, it is likely that they are stellar lines rather than interstellar lines. However,

this method failed as it flagged weak catalogue bands in the line-rich targets, where a different stellar line fell within the 0.3Å window in each observation. Therefore, a new method was tested which compared the flux at every wavelength across all observations, and would run the same checks on both the barycentric corrected and the original, pre-barycentric corrected files, as telluric lines will vary after barycentric corrections. However, this also failed as it produced spikes in the Hβ absorption bands, failed to remove the Na D lines in three of the six targets and only partially removed the telluric O_2 lines. Therefore, I decided to use a simple cut method where the Na D lines (5880Å–5905Å) and telluric O_2 (6274Å–6332Å) were set to a normalised value of 1, similar to the method in Simon et al., 2026.

This produced the following plots.

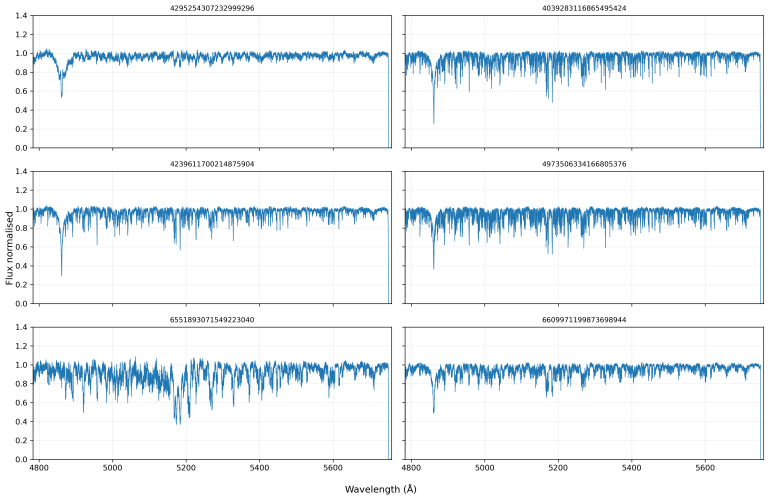


Figure 21: Grid plot of all spectra in the blue chip, barycentric and normalised with cosmic rays removed.

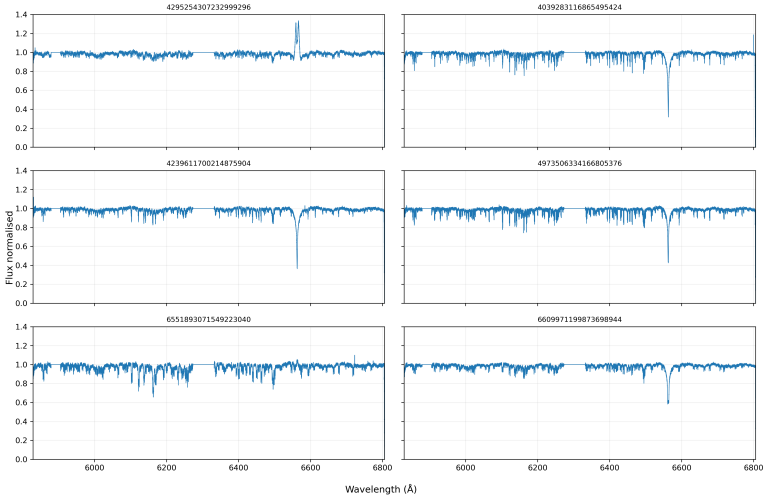


Figure 22: Grid plot of all spectra in the red chip, barycentric and normalised with cosmic rays removed, Na D and telluric O_2 lines removed.

Assumption six is intended to be handled next week.
Code: Added `error_pdf`, `cosmic_rays`, `line_marking` and `normalised_tagged` functions.
Files produced: PDF figure, spectral line-flagged figures.

Weekly summary

This week I started the project by reading two of the four introductory papers (El-Badry & Rix, 2022 and Simon et al., 2026). Using the methods from these two papers, I wrote a python script to identify 23 candidate binaries from the Gaia DR3 database and download their data for FEROS and UVES from the ESO archive, using ADQL. Then, choosing

UVES as the primary dataset, I processed the 6 UVES candidates through barycentric corrections and normalised their spectra, progressing from a min-max approach to a two-pass polynomial fit and finally a sigma-clipped Chebyshev fit. I also identified Fraunhofer lines in the spectra, along with other interesting features, particularly the Hα emission in 4295254307232999296, interstellar Na D and the telluric O_2 band. My meeting with Dr. Seeburger set out the cross-correlation method, the tools to use (TheJoker and his disentangling script) and the scope of the project. I also recreated the Gaia population plots from Simon et al. (2026), finding and fixing a query error in the process. Finally, I began to attempt some preparation for FFT cross-correlation, using the method from David et al., 2014. Next week I plan to satisfy assumption 6, improve the work to fully satisfy all the assumptions and begin to work toward cross-correlation against synthetic spectrum templates.

Plan status: Met plan to produce and inspect spectra of targets. Done some extra work in the cross-correlation test run preparation.

Required packages: astropy, astroquery, matplotlib, numpy, pandas, pyvo, requests and scipy.

Plots produced: raw, barycentric and normalised spectra of all observations (individual, combined, grid), semi-amplitude against period, significance against goodness-of-fit, CMD, Secondary mass against primary mass, noise PDF for all observations.

References: El-Badry & Rix (2022), Simon et al. (2026), Bagnulo et al. (2003), Wang et al. (2025), El-Badry et al. (2024), Poznanski et al. (2012), Sansonetti et al. (2010), Wang et al. (2022), David et al. (2014), Hobbs et al. (2008), FEROS-II User Manual, Gridihar (2010), Wikipedia, Fraunhofer (stand in for Meroni et al., 2010).

Tier 2 AI Policy:

I used Claude AI for bug fixing and planning code I critically evaluated all outputs and remain responsible for the submitted work.